

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo- and Active-Comparator-Controlled Clinical Study of Adjuvant V940 (mRNA-4157) Plus Pembrolizumab Versus Adjuvant Placebo Plus Pembrolizumab in Participants With High-Risk Stage II-IV Melanoma (INTerpath-001) **Short Title/Acronym:** A Clinical Study of Intismeran Autogene (V940) Plus Pembrolizumab in People With High-Risk Melanoma (V940-001) **Protocol Number:** V940-001 **NCT Number:** NCT05933577 **Posting ID:** 130786206856368179 **Sponsor Name:** Merck **Investigational Product:** V940 (mRNA-4157 / intismeran autogene) + Pembrolizumab (Keytruda, ATC Code: L01FF02) **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of adjuvant pembrolizumab plus V940 versus pembrolizumab plus placebo in improving Recurrence-Free Survival (RFS) in patients with completely resected high-risk stage II-IV melanoma.

Secondary Objectives: To evaluate Distant Metastasis-Free Survival (DMFS), Overall Survival (OS), the number of participants who experience an adverse event (AE) to determine safety, and quality of life (QoL).

2.2 Background & Rationale The purpose of this study is to learn if intismeran autogene which is an individualized neoantigen therapy (INT; formerly called messenger ribonucleic acid [mRNA]-4157) with pembrolizumab (MK-3475) is safe and prevents cancer from returning in people with high-risk melanoma. Researchers want to know if intismeran autogene with pembrolizumab is better than receiving pembrolizumab alone at preventing the cancer from returning. While adjuvant pembrolizumab has improved RFS and DMFS in patients with high-risk melanoma, many patients still experience disease recurrence. This Phase 3 study (INTerpath-001) aims to confirm previous findings, evaluating whether targeting a patient's unique tumor mutational signature with V940 can stimulate a stronger specific T-cell immune response and prevent cancer recurrence more effectively than the current standard of care.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and active-comparator-controlled **Blinding:**

Double-blind **Randomization:** 2:1 (V940 + pembrolizumab vs. placebo + pembrolizumab) **Trial Status:** ACTIVE_NOT_RECRUITING

3.2 Planned Enrollment & Duration Target **SN\$:** Approximately 1,089 patients **Number of Sites:** >165 global sites **Estimated Study Start:** 19-Jul-2023 **Estimated Primary Completion:** 26-Oct-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 2. Has surgically resected and histologically/pathologically confirmed diagnosis of cutaneous melanoma Stage IIB or IIC (pathologic or clinical), III, or IV per AJCC 8th edition. 3. Has not received any prior systemic therapy for their melanoma beyond surgical resection. 4. No more than 13 weeks have passed between the final surgical resection that rendered the participant disease-free and the first dose of pembrolizumab. 5. Is disease-free at the time of providing documented consent for the study. 6. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on anti-retroviral therapy (ART).

4.2 Exclusion Criteria 1. Has ocular or mucosal melanoma. 2. Has past or current in-transit metastases, satellitosis, or cancer that has spread to other parts of the body and cannot be removed with surgery. 3. Has received prior cancer therapy or another cancer vaccine. 4. Has heart failure within the past 6 months. 5. Has another known cancer that has spread to other parts of the body or has required treatment within the past 3 years. 6. Has severe reaction to study medications or any of their substance used to prepare a drug. 7. Have not recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - V940 (mRNA-4157 / intismeran autogene) 1 mg OR Placebo administered once every 3 weeks for up to 9 doses. - Pembrolizumab (Keytruda, ATC Code: L01FF02) 400 mg administered once every 6 weeks for up to 9 cycles. **Route of Administration:** Intramuscular (IM) for V940/Placebo; Intravenous (IV) for pembrolizumab. **Duration of Treatment:** Up to approximately 1 year (56 weeks), or until disease recurrence or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for adverse events and pre-existing co-morbidities. **Prohibited:** Any other systemic anti-cancer therapies, live-attenuated vaccines, or systemic immunosuppressive corticosteroids above physiological doses.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue sample provision for sequencing, Blood sample
Baseline	Day 0	Randomization, First Dose of pembrolizumab, First Dose of V940/Placebo (as soon as manufactured)
Interim	Q3W & Q6W	V940/Placebo administered Q3W; Pembrolizumab administered Q6W. Safety Labs, Efficacy Assessment, QoL Questionnaires
End of Study	Week 56	Final Treatment Visit, Exit Interview, Transition to long-term survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Recurrence-Free Survival (RFS) - defined as the length of time from when the participant starts the study until either the cancer comes back, or the cancer spreads as assessed by the investigator, or death due to any cause. **Measurement Method:** Investigator assessment using standard radiological imaging and clinical evaluation.

7.2 Secondary & Exploratory Endpoints - Distant Metastasis-Free Survival (DMFS) - Overall Survival (OS) - Number of Participants Who Experience an Adverse Event (AE) - Change from Baseline in the EORTC QLQ-C30 Role Functioning Score

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient interviews, physical exams, and lab assessments prior to each dosing cycle. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee to evaluate ongoing safety and efficacy parameters.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary and secondary efficacy endpoints. Safety Set for all safety and tolerability evaluations. **Sample Size Justification:** \$N=1,089\$ randomized 2:1 to provide adequate statistical power ($\alpha = 0.05$) to detect a clinically meaningful difference in RFS between the combination therapy and standard-of-care pembrolizumab monotherapy. **Handling of Missing Data:** Censoring applied for time-to-event outcomes (RFS, DMFS, OS) per standard Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence, drug accountability, and data integrity. **Audit Trail:** Continuous tracking via eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{V940-001}} **Posting ID:** {{130786206856368179}}
Registry Number: {{NCT05933577}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{A Clinical Study of Intismeran Autogene (V940) Plus Pembrolizumab in People With High-Risk Melanoma (V940-001)}} **Official Regulatory Title:** {{A Phase 3, Randomized, Double-Blind, Placebo- and Active-Comparator-Controlled Clinical Study of Adjuvant V940 (mRNA-4157) Plus Pembrolizumab Versus Adjuvant Placebo Plus Pembrolizumab in Participants With High-Risk Stage II-IV Melanoma (INTERpath-001)}}}

12. Clinical Framework (Melanoma Specific)

Primary Condition: {{Malignant Melanoma / High-Risk Stage II-IV Melanoma}} **Disease Terminology Codes:** {{MeSH: D008545 | SNOMED: 372244006 | HPO: HP:0002861}} **Diagnosis Threshold:** {{Surgically resected and histologically/pathologically confirmed diagnosis of Stage IIB, IIC, III, or IV cutaneous melanoma}} **Medical Methodology:** {{Adjuvant Immunotherapy and Individualized Neoantigen Therapy (INT)}} **Optimization Target:** {{Recurrence-Free Survival (RFS) improvement}}

13. Study Procedures & Methodology

Management Pathway: {{Adjuvant treatment post-complete resection}} **Education Protocol (HCP):** {{Training on mRNA vaccine administration, specialized handling, and immune-mediated adverse event management}} **Education Protocol (Patient):** {{Onsite training/self-learning regarding immunotherapy side effects and reporting requirements}} **Quality Audit Mechanism:** {{Patient adherence monitoring and AE tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 56 weeks of treatment, with long-term survival follow-up}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W) for V940/placebo dosing, and every 6 weeks (Q6W) for pembrolizumab}} **Remote Monitoring Frequency:** {{Between onsite

visits as required for safety monitoring}} **Checklist Review**
Interval: {{Routine imaging for disease recurrence every 12 weeks (Q12W)}}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				